

FINAL REPORT

**Financial Engineering Capstone:
QWIM Regime-based Investing with
LLMs**

Group 4

Submitted by

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Abstract

This project develops a regime-aware portfolio management framework that integrates market-state detection, bubble-risk estimation, out-of-sample signal generation, and mean-variance portfolio construction. We evaluate three complementary model families: Hidden Markov Models (HMM), Statistical Jump Models (SJM), and Log-Periodic Power Law Singularity (LPPLS) models. HMM and SJM are implemented under both time-domain and frequency-domain feature representations using ETF, volatility, and macroeconomic variables, while LPPLS generates asset-level bubble-confidence signals from daily ETF log prices. All model-driven strategies are evaluated against comparable StaticMVO benchmarks using the same asset universe, constraints, rebalance frequency, and performance metrics.

The empirical results show that statistical regime identification does not necessarily translate into superior portfolio performance. HMM produces interpretable and persistent latent states, especially in the frequency-domain specification, but both HMM-conditioned portfolios underperform their StaticMVO benchmarks. SJM provides more selective regime signals: the time-domain version improves maximum drawdown, while the frequency-domain version improves annualized return, but neither dominates StaticMVO across all risk-adjusted metrics. LPPLS delivers the clearest portfolio benefit in the current backtest, improving return, volatility, Sharpe ratio, Sortino ratio, Calmar ratio, and tail-risk measures with only a modest increase in turnover. Overall, the findings support a layered interpretation in which HMM and SJM are useful for regime monitoring and state-based analysis, while LPPLS is most effective as a sparse defensive overlay for portfolio risk management.

1 Introduction

Financial markets do not evolve under a single stationary environment. Instead, they experience persistent periods characterized by distinct patterns of returns, volatility, correlations, and macroeconomic conditions. These periods, commonly referred to as market regimes, are associated with different phases of the business cycle and investor sentiment. Major historical episodes such as the Dot-Com Bubble, the Global Financial Crisis of 2008, the COVID-19 market collapse, and the inflation-driven market adjustment beginning in 2022 illustrate how dramatically market behavior can change across regimes. Because assets respond differently to changing economic conditions, identifying and adapting to these regime shifts has become a central problem in modern portfolio management.

Traditional portfolio construction methods typically assume stable relationships among assets and rely on historical estimates of expected returns and risk. However, substantial evidence suggests that these relationships vary significantly across market environments. As a result, investment strategies that ignore regime changes may suffer from deteriorating performance, particularly during periods of heightened uncertainty and market stress. Consequently, an increasing body of literature has explored regime-aware investment frameworks in which market states are first identified and subsequently incorporated into forecasting and portfolio allocation decisions.

This project, inspired by the framework proposed by Shu et al. (2025), develops a complete regime-aware portfolio management framework that integrates feature preprocessing, market-state or bubble-signal estimation, out-of-sample signal generation, and portfolio construction. We evaluate three complementary model families: Hidden Markov Models (HMM), Statistical Jump Models (SJM), and Log-Periodic Power Law Singularity (LPPLS) models. HMM and SJM are used to identify and forecast broad market regimes, while LPPLS is used to detect asset-level bubble pressure and acts as a crash-risk overlay. To examine whether frequency-specific information improves performance, HMM and SJM are implemented under both time-domain and frequency-domain feature representations. All model-driven strategies are evaluated against comparable StaticMVO benchmarks using the same asset universe, constraints, rebalance frequency, and performance metrics.

The empirical results show that statistical regime detection does not automatically translate into superior portfolio performance. HMM produces interpretable latent states, especially in the frequency-domain specification, but both HMM-conditioned portfolios underperform their StaticMVO benchmarks. SJM provides more selective regime signals: the time-domain version improves maximum drawdown, while the frequency-domain version improves annualized return, but neither specification dominates StaticMVO across all risk-adjusted metrics. LPPLS provides the clearest portfolio benefit in the current backtest, improving return, volatility, risk-adjusted performance, and tail-risk measures with only a moderate increase in turnover. Overall, the project suggests that regime-aware signals are most useful when interpreted as complementary risk-management tools rather than as uniformly superior sources of portfolio alpha.

2 Literature Review

2.1 Objective and Summary

Market regime detection has become an important area of research in quantitative finance because financial markets exhibit persistent changes in returns, volatility, correlations, and macroeconomic conditions. Identifying these changes is directly relevant to asset allocation, risk management, and portfolio construction, since investment strategies designed under the assumption of a single stationary market environment may perform poorly when underlying market conditions shift. Consequently, a growing body of literature has developed statistical and structural methods for identifying market regimes, forecasting changes in market conditions, and incorporating such information into investment decisions. This review summarizes the main methodological foundations that motivate the models selected in this project.

Existing approaches to market-state analysis can be organized into three complementary strands. The first consists of probabilistic latent-state models, represented in this study by Hidden Markov Models (HMM), which model market regimes as unobservable states evolving over time. The second consists of clustering-based approaches with explicit temporal regularization, represented by Statistical Jump Models (SJM), which identify persistent market states while penalizing excessive transitions between regimes. The third focuses on structural models of market instability, represented by the Log-Periodic Power Law Singularity (LPPLS) framework, which analyzes speculative bubbles and the buildup of crash risk rather than classifying observations into discrete macro-financial regimes.

These three strands address related but distinct aspects of market dynamics. HMM provide an interpretable probabilistic framework and serve as a natural benchmark for temporal regime detection. SJM offer an alternative approach that emphasizes regime persistence, flexible feature construction, and subsequent supervised forecasting of the estimated regime labels. LPPLS complements these regime-classification methods by identifying asset-level price dynamics associated with speculative bubbles and potential market corrections. Taken together, the three methodologies provide a broader framework for analyzing both persistent market conditions and periods of heightened instability.

The following sections review prior work on each methodology, with particular attention to theoretical foundations, empirical applications, advantages and limitations, and suitability for integration into a regime-aware portfolio management framework. The review also considers how regime labels, regime probabilities, and crash-risk signals can be translated into portfolio decisions. This provides the methodological foundation for the empirical design used in the present study, where HMM, SJM, and LPPLS variants are evaluated against a common StaticMVO benchmark under comparable portfolio-construction assumptions.

2.2 Review of Hidden Markov Models (HMM)

Hidden Markov Models belong to the broader family of regime-switching models used to represent time series whose behavior changes across unobservable states. Hamilton (1989) introduced a regime-switching framework in which persistent changes in macroeconomic time series are represented by a latent Markov state process. Under this framework, the observed variables are generated from state-dependent distributions, while the latent state evolves according to a transition probability matrix. This structure is particularly

relevant to financial markets because changes in returns, volatility, correlations, and other market characteristics are often persistent rather than independent from one period to the next. Ang and Timmermann (2012) review how regime-switching models can reproduce several important properties of financial returns, including heteroskedasticity, skewness, fat tails, and time-varying cross-asset correlations.

An important strand of the literature examines the implications of regime changes for portfolio allocation. Ang and Bekaert (2002) model international equity returns under regimes with different expected returns, volatilities, and correlations, and demonstrate that regime-dependent investment opportunities can affect the value of international diversification and the optimal allocation between risky and conditionally risk-free assets. Similarly, Ang and Bekaert (2004) find that high-volatility regimes frequently coincide with bear markets and elevated cross-market correlations. Their results indicate that the economic value of regime information depends on the investment universe and may be greater when investors can allocate across equities, bonds, and cash rather than equities alone. These studies establish that regime identification is relevant not only for describing historical market behavior, but also for portfolio construction.

Later studies develop more operational regime-based allocation systems. Nystrup et al. (2018) examine whether dynamic allocation based on inferred market regimes can improve long-term results relative to a diversified static 60/40 portfolio. Their analysis highlights a central trade-off: responding to regime changes may improve portfolio performance, but aggressive regime-based positions may reduce diversification and increase concentration. The study therefore suggests that the value of a regime model should be assessed through portfolio-level outcomes and realistic benchmarks rather than through the statistical fit of the regime model alone.

The selection and treatment of model inputs constitute another important issue in HMM applications. Financial and macroeconomic datasets may contain a large number of correlated or uninformative variables, which can weaken regime identification and increase estimation uncertainty. Fons et al. (2021) address this problem using a Feature Saliency Hidden Markov Model that jointly performs feature selection and regime estimation. The authors use the resulting market states in a dynamic smart-beta allocation system and report improved risk-adjusted performance relative to portfolios based on an HMM using the full feature set. Their results support the use of HMM-based probabilities in dynamic allocation while also showing that careful feature selection may be important when the candidate information set is large.

Despite their interpretability, standard HMM have several limitations. Estimation commonly relies on maximum-likelihood methods such as the expectation-maximization algorithm, which may be sensitive to initialization and local optima. The inferred regimes may also depend on the assumed number of states and the selected state-conditional distributions. Moreover, a conventional first-order HMM may produce more frequent state changes than are economically plausible for medium- and long-horizon investment decisions. Nystrup et al. (2020) address this issue by introducing a jump penalty that provides direct control over the frequency of state transitions. Their results show that persistence regularization can reduce sensitivity to initialization and improve robustness when the state-conditional distributions are misspecified. This extension also provides a methodological link between conventional HMM and the Statistical Jump Models considered elsewhere in this project.

A related question is whether greater model complexity produces greater economic value. Hidden Semi-Markov Models relax the standard HMM duration structure by

modeling regime durations more explicitly. However, Baitinger and Hoch (2024) find that improvements in in-sample model fit from the more complex HSMM do not consistently translate into superior out-of-sample allocation performance. Their findings suggest that the number of states and the complexity of the duration model should be chosen with reference to out-of-sample investment outcomes rather than statistical fit alone.

Taken together, the literature supports HMM as an interpretable probabilistic framework for identifying persistent market states and translating regime uncertainty into portfolio decisions. At the same time, it emphasizes the need to control model complexity, limit the feature set, assess sensitivity to initialization and state count, and distinguish between retrospective regime identification and information that would have been available in real time. These considerations motivate the parsimonious Gaussian HMM and walk-forward implementation described in the methodology section. The HMM serves as a probabilistic temporal benchmark, while its results can be compared with the stronger transition regularization imposed by the Statistical Jump Model and evaluated through both regime stability and portfolio-level performance.

2.3 Review on Statistical Jump Models (SJM)

Statistical Jump Models (SJM) is proposed as another promising way of performing market detection. Traditional regime detection methods have largely been dominated by Hidden Markov Models (HMM), which assume that observations are generated by latent states following a Markov chain. Shu (2025) points out that existing models often suffer from model misspecification and estimation difficulties arising from latent variables and restrictive assumptions regarding state-conditional distributions. Furthermore, the non-convex optimization involved in likelihood estimation can lead to unstable results, especially in high-dimensional and noisy financial environments. Statistical Jump Models provide an attractive alternative by avoiding explicit probabilistic assumptions and instead treating regime identification as a clustering problem with temporal dependence. Specifically, SJM partition observations into clusters while imposing a jump penalty on state transitions. This regularization mechanism encourages persistence in the latent state sequence and prevents excessive switching between regimes, thereby producing market states that are more consistent with the long-lasting nature of economic cycles. According to Shu (2025), the jump penalty reflects the empirical observation that abrupt changes in market conditions are rare and that market regimes exhibit strong persistence over time.

Another important advantage of Statistical Jump Models lies in their flexibility regarding feature selection. SJM can accommodate a broad variety of features and data transformations. Consequently, the model can naturally incorporate macroeconomic indicators together with financial variables. Based on the studies of Möller and Hellgren (2025) and Xiong et al. (2026), a practical implementation of the Statistical Jump Model should employ a combination of macroeconomic and market variables. Macroeconomic information may be obtained from the FRED-MD database, which contains indicators describing output and income, labor markets, housing activity, money and credit conditions, interest rates, and inflation. Such variables provide information regarding the state of the business cycle and are widely used in tactical asset allocation studies. In addition, financial variables extracted from historical asset returns can be incorporated, including rolling returns, realized volatility, drawdowns, moving averages, correlations, yield spreads, and commodity prices. Möller and Hellgren (2025) apply the methodology to U.S. macroeconomic and financial data over the period 2003–2025 and demonstrate that

the resulting regimes exhibit clear differences in volatility, correlations, and drawdown characteristics. Similarly, Xiong et al. (2026) show that cross-asset return and volatility features can be used to construct asset-specific regime signals that remain robust even when auxiliary macroeconomic variables are excluded.

From the perspective of practical implementation, SJM are particularly attractive because existing literature demonstrates that they are directly compatible with the remaining stages of the proposed pipeline, including forecasting and portfolio optimization. Shu et al. (2025) propose a hybrid framework in which the Statistical Jump Model first generates regime labels through unsupervised learning, after which a supervised gradient-boosted decision tree model predicts future regimes. The regime forecasts are subsequently incorporated into Markowitz mean-variance optimization to determine portfolio weights. Their empirical study considers twelve representative assets, including U.S. large-cap, mid-cap and small-cap equities, developed-market equities, emerging-market equities, aggregate bonds, long-term Treasury bonds, high-yield bonds, corporate bonds, REITs, commodities, and gold. Using daily data from 1991 to 2023, the authors demonstrate that regime-aware portfolios consistently outperform traditional minimum-variance, mean-variance, and equally weighted portfolios. More recently, Xiong et al. (2026) extend this framework by incorporating equal-weight and risk-parity allocations and introducing gold as a defensive asset when all risky assets are classified as bearish. Their results show improved Sharpe ratios, Sortino ratios, and drawdown characteristics across multiple crises, including the Global Financial Crisis and the COVID-19 pandemic. These findings suggest that Statistical Jump Models should not be viewed as isolated clustering algorithms, but rather as central components of an end-to-end portfolio management architecture.

The strongest evidence supporting Statistical Jump Models comes from empirical performance studies. Shu et al. (2024a) investigate a regime-switching investment strategy designed to mitigate downside risk by reducing market exposure during unfavorable market conditions. Using daily data from major equity indices in the United States, Germany, and Japan over the period 1990–2023, and accounting for transaction costs and trading delays, the authors compare Statistical Jump Models with Hidden Markov Models and buy-and-hold strategies. Their results indicate that the Statistical Jump Model consistently delivers lower volatility, lower maximum drawdowns, and higher risk-adjusted returns. Importantly, the persistence induced by the jump penalty results in more interpretable regimes and greater robustness to implementation delays. Möller and Hellgren (2025) report similar findings using U.S. macroeconomic and financial data, showing that a regime-aware portfolio based on Jump Models achieved a Sharpe ratio of 0.68 compared with 0.54 for HMM. Collectively, these empirical studies demonstrate that Statistical Jump Models provide tangible economic benefits rather than merely improving statistical classification accuracy.

2.4 Review on Log-Periodic Power Law Singularity (LPPLS)

As we mentioned in reviewing the previous literature surrounding HMM and SJM, we find that traditional regime detection methods are designed primarily to identify the prevailing market state *ex post*. In particular, they do not explicitly address speculative bubbles and the crashes that frequently follow them. The Log-Periodic Power Law Singularity (LPPLS) framework, originally developed by Sornette and collaborators, addresses this complementary problem. Rather than partitioning historical observations into discrete

states, LPPLS models the evolution of asset prices during a bubble as a faster-than-exponential power-law growth process modulated by log-periodic oscillations converging toward a critical time. The estimated critical time, together with the calibrated parameters and filter conditions, provides a quantitative indication of how close the market is to an endogenous phase transition. Within the present project, LPPLS therefore serves as a risk-overlay model operating in parallel with the regime-detection layer and feeding its outputs into the same portfolio construction stage.

The theoretical foundation of LPPLS is more structural than that of purely statistical approaches. The model combines rational expectation bubble theory with concepts from the statistical physics of critical phenomena and interprets crashes not as exogenous shocks but as endogenous consequences of self-reinforcing imitation among heterogeneous investors. This framework leads to falsifiable structural assumptions and a relatively small number of economically interpretable parameters, making the model particularly suitable for client-facing explanations. Shu and Song (2024) provide a recent methodological review of LPPLS and a detailed application to the S&P 500 during the period 2020–2022. Their study discusses the standard nonlinear least-squares calibration procedure, the use of metaheuristics such as CMA-ES to avoid local optima, and the canonical filter conditions on the power-law exponent and log-periodic frequency that distinguish genuine bubble signatures from statistical artifacts. Filimonov et al. (2017) further develop inference procedures and interval forecasting for the critical time through a modified profile likelihood approach, allowing uncertainty surrounding the crash date to be quantified.

The empirical record of LPPLS is unusually extensive for a quantitative bubble-detection model. Zhang et al. (2016) investigate LPPLS indicators across two centuries of S&P 500 data and provide evidence that the methodology generalizes beyond the regimes used for calibration. Sornette et al. (2015) report a successful real-time prediction of the 2015 Shanghai stock market bubble and its subsequent collapse, representing one of the few publicly documented out-of-sample bubble forecasts. Wheatley et al. (2018) and Yao and Li (2021) extend the methodology to cryptocurrency markets and combine LPPLS with complementary diagnostics, demonstrating that the approach remains effective despite the structural differences between traditional and digital assets. Furthermore, the Financial Crisis Observatory at ETH Zürich has maintained a continuous real-time LPPLS deployment since 2008, providing one of the longest-running external validation records for any bubble-detection methodology.

An important feature of the present project is the incorporation of Large Language Models into the regime and risk analysis framework. LPPLS is particularly suitable for such integration because its outputs admit intuitive textual interpretations. When the model identifies a developing bubble, the LLM layer may retrieve and summarize contemporaneous financial news, central bank communications, earnings-call transcripts, and geopolitical developments in order to assess whether the quantitative signal is supported by a coherent narrative of excessive optimism, tightening monetary policy, or sentiment-driven speculation. Koa et al. (2024) propose a similar architecture based on temporal relational reasoning with large language models and show that incorporating narrative information can improve both the precision of crash detection and the interpretability of the resulting explanations. Such a narrative verification layer helps mitigate one of the principal weaknesses of purely statistical bubble detectors, namely their tendency to generate false positives during periods of sustained but justified price appreciation.

From the perspective of portfolio construction, LPPLS signals enter the same mean-variance optimization framework employed elsewhere in the project, although they serve

as a defensive overlay rather than a regime-conditioning variable. Gerlach et al. (2020) provide a direct precedent by incorporating a modified Kreuser–Sornette bubble model into a crash-sensitive Kelly strategy and demonstrate that LPPLS-based de-risking can substantially improve drawdown characteristics without sacrificing long-run expected returns. In the proposed framework, an asset whose bubble probability exceeds a calibrated threshold will receive a reduced portfolio weight. Moreover, the threshold itself may depend on client-specific risk preferences and investment horizons. Investors with shorter horizons or greater sensitivity to losses may trigger defensive adjustments at lower bubble probabilities, operationalizing the asymmetric loss aversion that motivates the overlay.

3 Data Description

3.1 Overview of Data

The proposed framework combines financial market data, volatility information, and macroeconomic indicators in order to describe the investment environment from multiple perspectives. The data are designed to support three related tasks: regime identification, regime forecasting, and portfolio construction. The empirical implementation uses a common investable universe across the Hidden Markov Model (HMM), Statistical Jump Model (SJM), and Log-Periodic Power Law Singularity (LPPLS) components, while allowing the feature representation to differ between the time-domain and frequency-domain experiments for HMM and SJM; the choice of data and features are supported by numerous literature discussed below.

The sample applied by models runs from April 17, 2006 to April 13, 2026. The discovery period ends on December 31, 2014, and the out-of-sample evaluation begins on January 1, 2015. The time-domain feature matrix contains 1,044 weekly observations and 58 variables. These consist of 28 ETF-based return and volatility features, 3 VIX-based volatility features, and 27 selected macroeconomic variables. The LPPLS component additionally uses daily adjusted ETF prices, which are transformed into log-price series for rolling bubble-confidence estimation.

Two feature domains are constructed for HMM and SJM. The time-domain specification uses the original weekly feature matrix after calendar alignment and stationarity transformations. The frequency-domain specification decomposes the weekly market variables and the monthly macroeconomic variables into multiple time-scale components using a Haar stationary wavelet transform. This design follows the intuition in the frequency-domain portfolio literature that aggregate predictors may hide useful information at different horizons, and that decomposed components can reveal predictive content that is obscured in the original series as per Faria and Verona (2024), Faria and Verona (2025).

3.2 Asset Universe and ETF-related Features

The financial component of the dataset consists of 14 exchange-traded funds (ETF) representing major asset classes, economic sectors, and geographic regions. These include SPY (broad U.S. equities), QQQ (technology and growth equities), EEM (emerging markets), EZU (Eurozone equities), XLE (energy), XLF (financials), XLK (technology), XLP (consumer staples), XLV (health care), XLB (materials), AGG (U.S. aggregate bonds), LQD (investment-grade corporate bonds), GLD (gold), and USO (oil).

The selection of these ETFs is motivated by the observation that different assets exhibit substantially different behavior across economic and financial regimes noted in Shu et al. (2024a). Cyclical sectors such as energy, financials, and materials tend to perform well during periods of economic expansion, while defensive sectors such as consumer staples and health care often demonstrate greater resilience during economic slowdowns. Fixed-income securities frequently benefit from declining interest rates during recessions, while commodities and precious metals may provide protection against inflationary shocks or heightened market uncertainty. Including assets with diverse economic sensitivities allows the regime detection models to identify meaningful differences in market conditions and provides the portfolio optimization stage with a sufficiently broad set.

Following Shu et al. (2025), return and volatility measures are used because they summarize market behavior relevant for regime identification. For HMM and SJM, the weekly ETF feature block contains two variables for each ETF: weekly return and weekly realized volatility. This produces 28 ETF-based variables. The same ETFs also form the portfolio construction universe. Therefore, the ETF data play a dual role: they provide explanatory features for regime recognition and constitute the assets over which the allocation model is optimized.

For LPPLS, the daily adjusted close prices of the same ETFs are used directly. Each price series is converted into log prices, and rolling LPPLS fits generate positive and negative bubble-confidence indicators for each asset. The LPPLS signal is therefore price-based rather than macro-based.

3.3 Volatility Features

In addition to asset returns, measures of market uncertainty are incorporated through the CBOE Volatility Index (VIX). The VIX is commonly interpreted as the market’s expectation of future volatility and is often referred to as the market’s “fear index.” Previous research like Shu et al. (2025) and Xiong et al. (2026) has shown that volatility measures contain information regarding investor sentiment, risk aversion, and transitions between market states.

Three features are constructed from daily VIX observations, including end-of-period VIX levels, average VIX levels, and measures of volatility variability over the observation period. These variables provide information beyond that contained in asset returns alone. During periods of elevated uncertainty, investors frequently alter risk exposures, correlations between assets increase, and portfolio diversification benefits may deteriorate. Especially, in the frequency-domain specification, the VIX variables are included in the weekly decomposition block together with the ETF return and volatility variables. Thus, VIX information is decomposed into short-horizon and lower-frequency components before entering HMM or SJM.

3.4 Macroeconomic Indicators Features

Macroeconomic information is obtained from the FRED-MD database, a widely used collection of U.S. economic indicators designed for empirical macroeconomic analysis previously applied in Oliveira et al. (2025) for macroeconomic regime detection. Following McCracken and Ng (2016), variables are transformed according to their prescribed transformation codes to improve stationarity and comparability across series. To balance information content, interpretability, and sample size considerations, a subset of econom-

ically meaningful indicators is selected rather than using the full set of over one hundred FRED-MD variables.

The selected variables cover several key dimensions of the economy, including consumption and income (RPI, W875RX1, DPCERA3M086SBEA), industrial production (INDPRO, IPFINAL, IPCONGD), labor market conditions (UNRATE, UEMPMEAN, CLAIMSx, PAYEMS), housing activity (HOUST, PERMIT), money and credit conditions (M2SL, BUSLOANS), interest rates and credit markets (FEDFUNDS, TB3MS, GS10, AAA, BAA), inflation (CPIAUCSL, PCEPI), investor sentiment (UMCSENTx), and exchange rates (TWEXAFEGSMTHx).

Together, these variables provide a broad representation of economic growth, employment, monetary policy, financial conditions, inflationary pressures, and investor expectations. Since market regimes are often associated with changes in these underlying economic conditions, incorporating macroeconomic indicators allows the regime detection models to capture information that may not be fully reflected in asset prices alone.

In addition, Yield Curve is constructed as difference between GS10 and TB3MS; Credit Spread as difference between BAA and AAA. The prior captures the slope of the term structure of interest rates and has historically been associated with future economic activity, while the latter measures perceived credit risk and financial stress. These derived variables complement the broader macroeconomic dataset and provide additional information regarding potential changes in economic and market conditions.

Macroeconomic variables are aligned according to their information release timing to avoid look-ahead bias. Depending on the final model specification, monthly observations may be retained directly or expanded to weekly frequency through forward-filling while preserving the information available to investors at each point in time.

3.5 Time-Domain Feature Matrix

The time-domain feature matrix is constructed at weekly frequency by joining the ETF, VIX, and macroeconomic features, which are discussed in the previous sections, on common weekly dates. After filtering to the model sample and dropping missing observations, the resulting matrix contains 58 variables, which include 28 ETF features, 3 VIX features, and 27 macroeconomic features mentioned.

This matrix is used directly by the time-domain HMM and SJM. In this specification, the models observe aggregate weekly features rather than explicit frequency components. The time-domain design therefore serves as the natural benchmark for testing whether frequency decomposition adds information beyond the original aligned feature set.

3.6 Frequency-Domain Feature Matrix

The frequency-domain specification is designed to expose horizon-specific information in the predictors. This is motivated by the observation that economic and financial variables may contain both short-run noise and lower-frequency cyclical information. Following the frequency-domain asset-allocation literature Faria and Verona (2024), Faria and Verona (2025), the project uses a Haar wavelet decomposition to separate each predictor into detail components and a smooth component.

For a generic time series X_t , the decomposition can be written as

$$X_t = \sum_{j=1}^J D_{j,t} + S_{J,t}, \quad (1)$$

where $D_{j,t}$ denotes the detail component at scale j , and $S_{J,t}$ denotes the smooth component at the final decomposition level. In the current implementation, the transform is implemented using the PyWavelets stationary wavelet transform with a Haar filter. This is an undecimated, MODWT-compatible implementation, which keeps each component aligned with the original time index.

The frequency-domain construction differs by source frequency. Weekly ETF and VIX variables are decomposed using level $J = 2$. This creates the following components:

$$\begin{aligned} D1 &: \text{approximately 2–4 weeks,} \\ D2 &: \text{approximately 4–8 weeks,} \\ S2 &: \text{approximately longer than 8 weeks.} \end{aligned} \quad (2)$$

There are 31 weekly variables in total, consisting of 28 ETF variables and 3 VIX variables. Therefore, the weekly frequency block contains 93 raw frequency-domain variables.

Monthly macroeconomic variables are decomposed using level $J = 4$. This creates the following components:

$$\begin{aligned} D1 &: \text{approximately 2–4 months,} \\ D2 &: \text{approximately 4–8 months,} \\ D3 &: \text{approximately 8–16 months,} \\ D4 &: \text{approximately 16–32 months,} \\ S4 &: \text{approximately longer than 32 months.} \end{aligned} \quad (3)$$

There are 27 macroeconomic variables, so the monthly macro frequency block contains 135 raw frequency-domain variables. Combining the weekly and monthly blocks gives a total of 228 raw frequency-domain features.

The frequency-domain features are constructed causally. For each weekly model date, the weekly SWT components are recomputed using only data available up to that date, and only the latest component values are retained. For monthly macroeconomic variables, the model applies a one-month publication lag before computing the available monthly history. This prevents the model from using macroeconomic information that would not yet have been available to investors at the rebalance date.

Because the raw frequency-domain feature matrix is high-dimensional, principal component analysis is applied within economically defined blocks. The blocks are defined by source frequency and wavelet component, namely `Weekly_D1`, `Weekly_D2`, `Weekly_S2`, `Monthly_D1`, `Monthly_D2`, `Monthly_D3`, `Monthly_D4`, and `Monthly_S4`.

Within each block, features are standardized and PCA retains enough components to explain at least 85 percent of the variance, subject to a minimum of one and a maximum of three retained components per block. In the current output, this produces up to 24 PCA features across the eight blocks. PCA is fit only within the discovery sample for model selection and refit only on historical data during walk-forward portfolio construction.

3.7 Data Usage Across HMM, SJM, and LPPLS

The HMM and SJM models are evaluated under both time-domain and frequency-domain data specifications. In the time-domain case, both models use the 58-variable weekly feature matrix. In the frequency-domain case, both models use the 228 raw wavelet components after blockwise PCA compression. The purpose is to test whether regime recognition and portfolio performance improve when the models are allowed to distinguish short-horizon market movements from slower macroeconomic cycles.

LPPLS uses the data differently. Its bubble-confidence indicators are estimated from daily ETF log prices and therefore do not use the VIX or FRED-MD feature matrix. As a result, the LPPLS signal itself is the same conceptual object in both the time-domain and frequency-domain experiments. The distinction between LPPLS time domain and LPPLS frequency domain arises from the StaticMVO benchmark and rebalance universe to which the overlay is applied. In the frequency-domain LPPLS experiment, the static benchmark weights are computed on the same frequency-domain-aligned weekly return dates used by HMM and SJM, while the LPPLS overlay continues to use the latest available asset-level bubble-confidence signal.

This design keeps the portfolio evaluation comparable across models. HMM and SJM differ in how they infer regimes; LPPLS differs by acting as an asset-level de-risking overlay. However, all three models are ultimately compared against a StaticMVO benchmark using the same asset universe, return conventions, performance metrics, and walk-forward evaluation period. The details will be discussed in the methodology section.

4 Methodology

4.1 Overall Framework

The methodology consists of four connected stages: feature preprocessing, market-state or bubble-signal estimation, out-of-sample signal generation, and portfolio construction. The objective is not only to identify statistically distinct market environments, but also to test whether such information improves portfolio outcomes relative to a regime-unaware benchmark, which in this case is vanilla Mean-variance optimization (MVO).

As described in Section 3, the project evaluates both time-domain and frequency-domain feature representations. The time-domain specification uses the original weekly ETF, VIX, and macroeconomic feature matrix. The frequency-domain specification uses the causally aligned wavelet-based feature representation constructed from weekly market variables and monthly macroeconomic variables. This section therefore focuses on how these features are used by the models, rather than repeating the construction of the frequency-domain variables.

Three model families are implemented. Hidden Markov Models (HMM) provide a probabilistic latent-state benchmark. Statistical Jump Models (SJM) provide a clustering-based regime detection method with explicit temporal regularization. Log-Periodic Power Law Singularity (LPPLS) models provide asset-level bubble-confidence indicators and serve as a crash-risk overlay rather than a discrete macro-regime classifier. The three approaches are intentionally complementary: HMM and SJM identify broad market regimes, while LPPLS identifies unstable price dynamics at the asset level.

The empirical design follows a strict discovery and out-of-sample structure. The discovery period ends on December 31, 2014 and is used for hyperparameter selection

and initial regime characterization. The out-of-sample evaluation begins on January 1, 2015. During the out-of-sample period, models are re-estimated using only information available before each rebalance date. Portfolio returns are realized beginning on the first trading day after the rebalance signal date, which avoids look-ahead bias.

4.2 Feature Preprocessing

Before HMM and SJM estimation, all input features are standardized using z-score normalization:

$$\tilde{x}_{i,t} = \frac{x_{i,t} - \mu_i}{\sigma_i}, \quad (4)$$

where μ_i and σ_i denote the sample mean and standard deviation of feature i estimated from the relevant training sample. In discovery-period model selection, the scaler is fit using only discovery-period observations. In walk-forward evaluation, the scaler is refit using only observations prior to the rebalance date.

For frequency-domain HMM and SJM specifications, the raw decomposed features described in Section 3 are further compressed using blockwise principal component analysis. The blocks are defined by source frequency and wavelet component, such as weekly detail components, weekly smooth components, monthly detail components, and monthly smooth components. Within each block, PCA retains enough components to explain at least 85 percent of the variance, subject to a minimum of one and a maximum of three principal components per block. PCA is always fit within the relevant training window and then applied to the current observation, so the dimension reduction step does not introduce future information.

This preprocessing choice serves two purposes. First, it makes features with different units and scales comparable. Second, in the frequency-domain specification, it reduces the dimensionality of the decomposed feature set while preserving the main common variation within each economically interpretable frequency block.

4.3 Hidden Markov Model

Hidden Markov Models are used as the probabilistic temporal benchmark for regime detection. HMM assume that observed market and macroeconomic variables are generated by an unobservable state process that evolves according to a first-order Markov chain. Let $z_t \in \{1, \dots, K\}$ denote the latent regime at time t , and let x_t denote the standardized feature vector. The transition probability from regime i to regime j is

$$P(z_t = j \mid z_{t-1} = i) = a_{ij}, \quad (5)$$

where $A = (a_{ij})$ is the transition matrix. Conditional on the latent regime, the feature vector follows a Gaussian emission distribution:

$$x_t \mid z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k). \quad (6)$$

The current implementation uses a Gaussian HMM with diagonal covariance matrices. This specification is deliberately parsimonious. Since both the time-domain and frequency-domain feature sets contain many correlated financial and macroeconomic variables, a full covariance HMM could become unstable and difficult to estimate reliably. A diagonal covariance HMM therefore provides a transparent probabilistic baseline while limiting overfitting.

Candidate state counts $K \in \{2, 3, 4\}$ were estimated separately for the time- and frequency-domain specifications using discovery-period observations only. Each candidate was estimated from five initializations, using seeds 42 through 46, and the fit with the highest log-likelihood was retained.

The HMM produces regime probabilities directly. At each out-of-sample rebalance date, the HMM is refit using only prior observations. The fitted model filters the information available up to the rebalance date and then computes the one-step-ahead regime probability:

$$\hat{p}_{t+1|t} = \alpha_t A, \quad (7)$$

where $\alpha_t = P(z_t \mid x_{\leq t})$ is the filtered state probability vector. The predicted regime is the state with the largest one-step-ahead probability. Thus, the implementation uses an argmax probability rule rather than requiring any regime probability to exceed 50 percent.

This HMM specification is consistent with the role of HMM in the regime-switching literature. Following Hamilton’s latent-state framework and subsequent asset-allocation applications, the HMM provides interpretable state probabilities, transition probabilities, and expected regime durations. It also serves as a natural benchmark against which the stronger persistence regularization of the SJM can be evaluated.

4.4 Statistical Jump Model

The Statistical Jump Model is implemented as a temporally regularized clustering method for market regime detection. Unlike the HMM, the SJM does not require a full probabilistic emission model. Instead, it partitions observations into clusters while penalizing excessive switching between regimes. Let $s_t \in \{1, \dots, K\}$ denote the regime assignment and let η_k denote the centroid of regime k in feature space. The SJM solves

$$\min_{\{s_t\}, \{\eta_k\}} \sum_{t=1}^T \|x_t - \eta_{s_t}\|^2 + \lambda \sum_{t=2}^T \mathbf{1}(s_t \neq s_{t-1}), \quad (8)$$

where λ is the jump penalty.

The first term measures within-regime dispersion, while the second term discourages unnecessary regime transitions. This structure follows the motivation in the Statistical Jump Model literature: financial regimes should be flexible enough to respond to changing market conditions, but persistent enough to avoid noisy over-switching. The jump penalty is therefore economically meaningful. A smaller λ permits more responsive regimes, while a larger λ produces smoother and more persistent regimes.

The project searches over $K \in \{2, 3, 4\}, \lambda \in \{10, 25, 50\}$. The model is initialized using K-means with 50 initializations, random seed 42, maximum iteration count 500, and convergence tolerance 10^{-4} . Given centroids, the optimal regime path is updated through a Viterbi-style dynamic programming step that incorporates the jump penalty. Centroids and labels are then updated iteratively until convergence.

Candidate SJM specifications are selected using the discovery period only. As with the HMM, eligible models must avoid empty regimes, maintain a non-negligible minimum occupancy, and avoid degenerate transition behavior. The final composite score rewards persistence, minimum occupancy, lower within-regime dispersion, and meaningful average regime duration. The current project outputs select $K = 2$ and $\lambda = 10$ for the main SJM specifications. This does not mean that regimes should switch frequently by design;

rather, it means that within the discovery-period selection rule, the more responsive jump-penalty setting provided the best balance between fit, persistence, and occupancy.

Because SJM does not produce probabilistic forecasts by itself, a supervised classifier is used after regime labels are estimated. Following the SJM forecasting approach in the literature, the project trains an XGBoost classifier to estimate one-step-ahead regime probabilities from the available feature vector. The implemented classifier uses 150 trees, maximum depth 3, learning rate 0.05, subsample ratio 0.85, column subsample ratio 0.85, random seed 42, and a multiclass soft-probability objective. At each rebalance date, the SJM labels and XGBoost classifier are re-estimated using only historical data. The predicted regime is the largest-probability regime, again using an argmax rule rather than a 50 percent threshold.

4.5 Log-Periodic Power Law Singularity Model

The Log-Periodic Power Law Singularity (LPPLS) model is used to detect asset-level bubble and crash-risk signals. Unlike the Hidden Markov Model (HMM) and the Statistical Jump Model (SJM), which classify the entire feature matrix into latent market regimes, LPPLS models the price dynamics of each individual asset during speculative bubble episodes. Following the LPPLS literature of Sornette and subsequent extensions (Shu and Song, 2024), the logarithm of the asset price is modeled as

$$\log P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \log(t_c - t) - \phi). \quad (9)$$

Here, t_c denotes the critical time associated with a potential regime transition or bubble termination, m is the power-law exponent governing faster-than-exponential growth, ω is the angular frequency of the log-periodic oscillation, and A , B , C , and ϕ are additional fitted parameters. In the common Filimonov–Sornette reparameterization shown by Filimonov and Sornette (2013), the linear coefficients can be analytically slaved to the nonlinear parameters (t_c, m, ω) , reducing the dimension of the nonlinear calibration problem and improving numerical stability.

In the current implementation, LPPLS is applied separately to each ETF using daily log prices. The model is estimated over rolling windows following the standard LPPLS calibration framework, using a window size of 252 trading days, a smallest window size of 126 trading days, an outer increment of 20 days, an inner increment of 10 days, a maximum of 12 calibration searches, and four parallel workers. These settings correspond approximately to a one-year estimation window, a half-year minimum fitting window, and a computationally feasible nested rolling-search procedure.

During calibration, only parameter estimates satisfying the standard LPPLS filter conditions are retained, including the admissible parameter ranges $m \in [0.01, 0.99]$ and $\omega \in [2, 25]$, following the filtering procedure commonly adopted in the LPPLS literature. The final output is a positive and negative bubble-confidence series for each ETF. The positive bubble-confidence series is subsequently incorporated into portfolio construction.

LPPLS enters the investment process as a quantitative crash-risk overlay applied to the StaticMVO allocation. Let $c_{i,t}$ denote the latest available positive bubble confidence for asset i at rebalance date t . The confidence value is transformed into a continuous de-risking multiplier

$$m(c_{i,t}) = 1 - (1 - \kappa) \frac{\sigma(a(c_{i,t} - \theta)) - \sigma(-a\theta)}{\sigma(a(1 - \theta)) - \sigma(-a\theta)}. \quad (10)$$

The logistic function is defined as

$$\sigma(u) = \frac{1}{1 + e^{-u}}. \quad (11)$$

The default parameters are set to $\theta = 0.5$, $a = 12$, and $\kappa = 0.3$. Under this specification, zero bubble confidence leaves the portfolio weight unchanged, while a bubble confidence of one reduces the risky exposure to 30% of its original allocation. Any residual weight removed through the overlay is allocated to a cash sleeve.

Recent developments have proposed further extensions of LPPLS. Filimonov et al. (2017) develop profile-likelihood inference for estimating confidence intervals of the critical time, while Nielsen, Sornette, and Raissi (2024) introduce Deep LPPLS, which replaces iterative nonlinear optimization with a neural-network approximation trained on synthetic LPPLS trajectories. In addition, Koa et al. (2024) demonstrate that Large Language Models can provide textual verification of bubble signals using financial news and corporate disclosures. These developments motivate future enhancements of the framework. Nevertheless, the current project adopts the classical rolling LPPLS implementation as a transparent and interpretable quantitative overlay. Its purpose is to reduce exposure when asset-level price dynamics exhibit elevated bubble confidence rather than to replace the HMM or SJM regime-classification layer.

4.6 Out-of-Sample Signal Generation

After feature preprocessing and model estimation, each model is converted into an implementable out-of-sample investment signal. This stage is deliberately separated from discovery-period regime identification. The discovery sample is used only for hyperparameter selection and initial model interpretation. All out-of-sample portfolio decisions are generated through a walk-forward procedure in which the relevant preprocessing transformations, model parameters, signals, and portfolio inputs are re-estimated using only information available at the rebalance date.

Let $\mathcal{F}_t$ denote the information set available at rebalance date t . For each monthly rebalance date in the out-of-sample period, model estimation uses observations strictly prior to t , while the feature vector available at t is used to generate the signal for the next holding period. Portfolio returns are measured beginning on the first trading day after the rebalance date. This timing convention prevents future information from entering either signal generation or realized portfolio-return measurement. The three model families generate different forms of out-of-sample signals. HMM produces one-step-ahead regime probabilities directly from its estimated latent-state structure. SJM produces historical regime labels that are subsequently forecast through the XGBoost classifier described above. LPPLS produces asset-specific bubble-confidence indicators rather than common market-regime probabilities.

4.6.1 HMM Signal Generation

At each rebalance date t , the HMM is re-estimated using only historical observations available before the portfolio decision. The fitted model filters the observations available through date t and propagates the filtered state probabilities one step ahead using the estimated transition matrix:

$$\hat{p}_{t+1|t}^{\text{HMM}} = \alpha_t A, \quad (12)$$

where $\alpha_t = P(z_t | x_{\leq t})$ is the filtered probability vector and A is the estimated transition matrix. When a discrete regime label is required, the predicted state is selected according to

$$\hat{z}_{t+1}^{\text{HMM}} = \arg \max_k \hat{p}_{k,t+1|t}^{\text{HMM}}. \quad (13)$$

The HMM therefore always provides a complete probability distribution across regimes. No individual state probability is required to exceed 50%; the largest-probability state is used only when a discrete point forecast is needed.

4.6.2 SJM Signal Generation

Because the SJM is a clustering model rather than a probabilistic forecasting model, its estimated regime labels must be converted into an out-of-sample prediction problem. At each rebalance date t , the SJM is re-estimated using only historical observations, producing a regime-label sequence $\{s_1, s_2, \dots, s_{t-1}\}$. These historical labels are then used to construct a one-step-ahead supervised learning problem in which each feature vector x_τ is mapped to the next-period regime label $s_{\tau+1}$, for $\tau < t$.

The XGBoost classifier described in the SJM subsection is trained only on observations available before the rebalance decision. Given the feature vector observed at date t , the classifier produces the probability vector

$$\hat{p}_{k,t+1|t}^{\text{SJM}} = P(s_{t+1} = k | x_t, \mathcal{F}_t), \quad k = 1, \dots, K. \quad (14)$$

When a discrete prediction is required, the forecast regime is

$$\hat{s}_{t+1}^{\text{SJM}} = \arg \max_k \hat{p}_{k,t+1|t}^{\text{SJM}}. \quad (15)$$

Both components of the SJM forecasting procedure are therefore estimated out of sample: the SJM generates historical training labels using only the expanding historical window, and the classifier maps the current feature vector into one-step-ahead regime probabilities.

4.6.3 LPPLS Signal Generation

LPPLS does not generate a common latent market regime and does not require a separate regime-forecasting classifier. Instead, it produces asset-level bubble-confidence indicators from rolling fits to the daily log-price history of each ETF. For asset i , the signal used at rebalance date t is the latest positive bubble-confidence estimate available within $\mathcal{F}_t$:

$$c_{i,t} = \text{LPPLSConf}_i^+(\tau_t), \quad (16)$$

where τ_t is the latest LPPLS estimation date available within $\mathcal{F}_t$. If no historical LPPLS estimate is available for an asset, its confidence value is set to zero. The resulting signal measures asset-specific bubble pressure and is subsequently used as a de-risking overlay rather than as an input to common market-regime classification.

4.7 From Signals to Portfolio Inputs

The out-of-sample signals are translated into portfolio inputs differently for the regime models and the LPPLS model. For HMM and SJM, historical weekly asset returns are

grouped according to the regimes estimated within the relevant training window. For each regime k , the model estimates a regime-specific expected return vector μ_k and covariance matrix Σ_k . Given the forecast regime-probability vector $\hat{p}_{t+1|t}$, the expected return vector used in portfolio construction is

$$\mu_t = \sum_{k=1}^K \hat{p}_{k,t+1|t} \mu_k. \quad (17)$$

The conditional covariance matrix is constructed as

$$\Sigma_t = \sum_{k=1}^K \hat{p}_{k,t+1|t} \Sigma_k + \sum_{k=1}^K \hat{p}_{k,t+1|t} (\mu_k - \mu_t)(\mu_k - \mu_t)^\top. \quad (18)$$

The first term captures expected within-regime covariance, while the second term captures uncertainty arising from differences across regime-specific expected returns. This formulation uses the full regime-probability distribution rather than only the most likely state. Portfolio inputs can therefore change smoothly even when the discrete predicted regime remains unchanged.

For LPPLS, the signal is not used to estimate regime-conditional return moments. Instead, the positive bubble confidence $c_{i,t}$ is converted into the continuous de-risking multiplier defined in the LPPLS subsection. The multiplier reduces the exposure of assets displaying elevated bubble confidence while leaving assets with negligible confidence largely unchanged.

4.8 Portfolio Construction

Portfolio construction is based on long-only mean-variance optimization. For HMM and SJM, the optimizer uses the probability-weighted expected return vector μ_t and covariance matrix Σ_t defined above. At each rebalance date, the portfolio solves

$$\max_w \quad w^\top \mu_t - \frac{\gamma}{2} w^\top \Sigma_t w, \quad (19)$$

subject to

$$\sum_{i=1}^N w_i = 1 \quad (20)$$

and

$$0 \leq w_i \leq 0.30, \quad i = 1, \dots, N. \quad (21)$$

The default risk-aversion parameter is $\gamma = 3$, and the maximum weight assigned to an individual risky asset is 30%. Covariance matrices are estimated using Ledoit–Wolf shrinkage when sufficient observations are available, improving numerical stability when some regime-specific samples are relatively small.

The benchmark strategy is StaticMVO. It uses the same asset universe, risk-aversion parameter, covariance-estimation procedure, long-only constraint, and maximum-weight constraint as the regime-aware strategies. However, it estimates expected returns and covariances without conditioning on HMM or SJM regime probabilities. StaticMVO therefore serves as the regime-unaware mean-variance benchmark.

For the LPPLS strategy, the StaticMVO allocation is computed first. Each risky asset weight is then multiplied by its asset-specific LPPLS de-risking multiplier. Let $w_{i,t}^{\text{Static}}$

denote the StaticMVO weight and let $m(c_{i,t})$ denote the multiplier defined previously. The adjusted risky-asset weight is

$$\tilde{w}_{i,t} = w_{i,t}^{\text{Static}} m(c_{i,t}). \quad (22)$$

The residual allocation is assigned to the cash sleeve:

$$w_t^{\text{cash}} = 1 - \sum_{i=1}^N \tilde{w}_{i,t}. \quad (23)$$

The resulting strategy is denoted LPPLSOverlayMVO.

4.9 Rebalancing and Walk-Forward Implementation

Given that we focus on mid-term to long-term investment per the QWIM requirement, rebalancing occurs monthly. The rebalance date is defined as the first available weekly model date in each calendar month during the out-of-sample period. This convention aligns the portfolio decision with the weekly feature calendar while maintaining a monthly allocation frequency suitable for medium-term tactical asset allocation.

At every rebalance date, the complete model-specific estimation and portfolio-construction procedure is repeated using an expanding historical window. The preprocessing transformations are refit, the relevant HMM or SJM specification is re-estimated, the out-of-sample signal is generated, the required return and covariance inputs are constructed, and the portfolio optimization is solved. For LPPLS, the latest bubble-confidence estimates available at the decision date are matched to the corresponding StaticMVO allocation and converted into asset-specific exposure adjustments.

The portfolio selected at date t becomes active on the first trading day after the rebalance date and remains in effect until the next rebalance. Daily realized strategy returns are therefore computed as

$$r_d^p = w_t^\top r_d, \quad (24)$$

where w_t is the portfolio selected at the most recent rebalance date and r_d is the vector of daily asset returns on trading day d within the associated holding period. This implementation ensures that the backtest represents an implementable sequence of real-time portfolio decisions rather than an ex-post regime-classification exercise.

4.10 Model Variants and Evaluation

The project evaluates six model variants across three model families and two feature domains: HMM-Time, HMM-Frequency, SJM-Time, SJM-Frequency, LPPLS-Time, and LPPLS-Frequency. The LPPLS signal itself does not methodologically differ across domains because it is estimated directly from daily ETF log prices.

HMM-Time and SJM-Time use the original time-domain weekly feature matrix. HMM-Frequency and SJM-Frequency use the frequency-domain feature representation after the blockwise PCA compression described previously.

The LPPLS signal itself is identical across the time-domain and frequency-domain variants because LPPLS is estimated directly from daily ETF log prices. The distinction between LPPLS-Time and LPPLS-Frequency arises from the underlying portfolio framework to which the LPPLS overlay is applied. LPPLS-Time is aligned with the time-domain portfolio implementation, while LPPLS-Frequency is aligned with the frequency-domain implementation and its corresponding rebalance calendar and baseline allocation.

The LPPLS designation therefore identifies the overlay environment rather than a different LPPLS estimation procedure.

The models are evaluated from both statistical and investment perspectives. For HMM and SJM, regime quality is assessed using regime occupancy, persistence, the number of transitions, average regime duration, transition matrices, and regime-specific return and risk characteristics.

One-step-ahead regime-forecasting performance is evaluated using accuracy, balanced accuracy, macro precision, macro recall, macro F1 score. Accuracy measures the overall proportion of correctly predicted regimes, while balanced accuracy and macro-averaged measures reduce the influence of unequal regime frequencies.

Portfolio performance is evaluated using daily realized returns. The reported metrics include annualized return, annualized volatility, Sharpe ratio, Sortino ratio, Calmar ratio, maximum drawdown, 95% value-at-risk, 95% conditional value-at-risk, and average turnover.

The primary economic comparison is the incremental performance of each model-driven strategy relative to StaticMVO. This comparison isolates whether regime conditioning or the LPPLS bubble-risk overlay adds value beyond an otherwise equivalent long-only mean-variance allocation.

5 Findings

5.1 Regime Identification Results

5.1.1 Hidden Markov Model Results

Under the reported two-state specification, both HMM variants identified nonempty and persistent latent regimes during the discovery period. Table 1 summarizes the principal regime diagnostics.

Table 1: Discovery-Period HMM Regime Characteristics

Specification	Regime 0 share	Transitions	Persistence	Avg. duration
HMM-Time	33.4%	34	0.925	13.0 weeks
HMM-Frequency	15.1%	10	0.974	34.8 weeks

The time-domain states exhibited economically distinguishable market characteristics. Regime 0 was associated with lower average weekly asset returns, higher realized volatility, a mean VIX level of approximately 30.9, and a substantially larger SPY drawdown. Regime 1 was associated with positive average returns, lower volatility, and a mean VIX level of approximately 16.2. We therefore refer to Regime 0 as the stress state and Regime 1 as the lower-volatility state. These descriptive labels were assigned after estimation and were not imposed on the HMM.

The frequency-domain HMM generated substantially longer-lasting state assignments. It recorded only 10 discovery-period transitions, compared with 34 for the time-domain specification, while its average regime duration increased from 13.0 to 34.8 weeks. Because the frequency-domain model was estimated from standardized wavelet components and principal components, the economic interpretation of its states was based on the cor-

responding date-aligned raw market observations. The signs of the principal-component scores were not interpreted directly.

Out-of-sample regime forecasting. Forecast performance was evaluated over 136 monthly rebalance dates. At each date, the HMM was re-estimated using only historically available observations and generated a one-step-ahead state-probability vector. Table 2 reports the principal out-of-sample classification metrics.

Table 2: Out-of-Sample HMM Forecast Performance

Specification	Accuracy	Balanced Accuracy	M. Precision	M. Recall	M. F1
HMM-Time	82.35%	90.84%	58.62%	90.84%	59.66%
HMM-Frequency	98.53%	96.43%	99.09%	96.43%	97.69%

The time-domain HMM correctly classified 112 of 136 one-step-ahead observations, corresponding to an accuracy of 82.35%. It correctly identified all 5 subsequently inferred Regime 0 observations and 107 of 131 Regime 1 observations. However, 24 Regime 1 observations were incorrectly classified as Regime 0. Balanced accuracy is therefore relatively high at 90.84%, because recall is 100% for Regime 0 and 81.68% for Regime 1. In contrast, macro precision and macro F1 are only 58.62% and 59.66%, respectively, because the precision of Regime 0 is only 17.24%. The time-domain model therefore detects the rare state sensitively but generates a substantial number of false Regime 0 signals.

The frequency-domain HMM correctly classified 134 of 136 observations. It identified 26 of the 28 subsequently inferred Regime 0 observations and all 108 Regime 1 observations. The two errors were Regime 0 observations classified as Regime 1; no Regime 1 observation was incorrectly classified as Regime 0. It therefore achieved 98.53% accuracy, 96.43% balanced accuracy, and 97.69% macro F1, substantially outperforming the time-domain specification under the reported two-state implementation.

The evaluation target is a subsequently inferred HMM state rather than a directly observed economic regime. Accordingly, these statistics measure the consistency with which the model forecasts its own latent state structure. They should not be interpreted as classification accuracy against objectively observed market-regime labels.

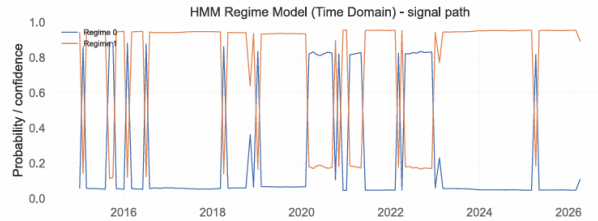


Figure 1: Out-of-sample HMM-Time one-step-ahead regime probabilities from 2015 to 2026. Regime 0 represents the state associated with lower mean SPY returns, while Regime 1 represents the higher-return state.

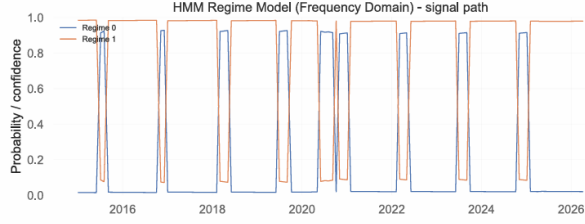


Figure 2: Out-of-sample HMM-Frequency one-step-ahead regime probabilities from 2015 to 2026. The frequency-domain specification produces more sharply separated and longer-lasting probability episodes than the time-domain specification.

Figures 1 and 2 illustrate the different probability dynamics of the two specifications. HMM-Time produces more short-lived probability reversals, whereas HMM-Frequency generally maintains its state assignments for longer intervals. This visual pattern is consistent with the greater discovery-period persistence and average regime duration reported for HMM-Frequency.

5.1.2 Statistical Jump Model Results

The out-of-sample regime-classification results for the Statistical Jump Model are reported as follows. Although the selected SJM specifications are chosen using the discovery period, the regime probabilities reported here are generated during the test period through the walk-forward procedure described in Section 4. At each monthly rebalance date, the SJM is re-estimated using only historical information, and an XGBoost classifier produces one-step-ahead regime probabilities. The displayed regime is the model’s out-of-sample predicted regime, defined by the largest predicted probability.

The SJM-Time and SJM-Frequency specifications both select a two-regime structure with jump penalty $\lambda = 10$. As shown in Table 3, in the out-of-sample period from January 2015 to April 2026, both models classify most months as Regime 1. However, SJM-Time assigns only 2 of 136 monthly rebalance dates to Regime 0, while SJM-Frequency assigns 6 of 136 dates to Regime 0. Regime 0 is therefore rare in both models, but appears more frequently under the frequency-domain representation.

Table 3: Out-of-Sample SJM Predicted Regime Distribution

Specification	Regime 0 Count	Regime 0 Share	Regime 1 Count	Regime 1 Share
SJM-Time	2	1.47%	134	98.53%
SJM-Frequency	6	4.41%	130	95.59%

The time-domain SJM, as shown in Figure 3, identifies Regime 0 only during the immediate COVID-19 market stress period. The two Regime 0 predictions occur on April 6, 2020 and May 4, 2020, with predicted Regime 0 probabilities of 77.6% and 72.7%, respectively. Outside this interval, the model assigns almost all observations to Regime 1 with high confidence. The average maximum predicted probability is 99.1%. These results indicate that SJM-Time associates Regime 0 primarily with abrupt short-horizon market stress.

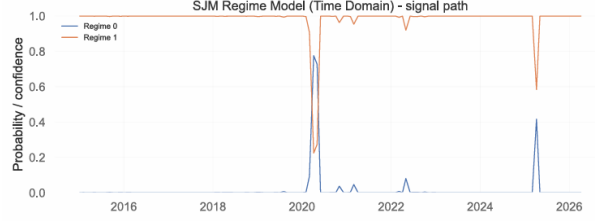


Figure 3: Out-of-sample SJM-Time regime probabilities. Regime 0 appears only during the COVID-19 market stress period, while Regime 1 dominates the remainder of the test period.

The frequency-domain SJM produces a broader but still selective Regime 0 classification. As shown in Table 4, Regime 0 appears on January 4, 2021; March 7, 2022; April 4, 2022; May 2, 2022; December 2, 2024; and January 6, 2025. The predicted Regime 0 probabilities range from 80.4% to 94.3%, indicating that these are not marginal classifications. The frequency-domain classifier is also generally confident, with an average maximum predicted probability of 99.0%.

Table 4: SJM-Frequency Out-of-Sample Regime 0 Dates

Date	Regime 0 Probability	Regime 1 Probability
2021-01-04	89.1%	10.9%
2022-03-07	80.4%	19.6%
2022-04-04	94.2%	5.8%
2022-05-02	83.8%	16.2%
2024-12-02	89.5%	10.5%
2025-01-06	94.3%	5.7%

The frequency-domain Regime 0 observations form three distinct episodes as shown in Figure 4: January 2021, March to May 2022, and December 2024 to January 2025. The first coincides with the post-pandemic reopening and reflation transition, the second with the combined geopolitical, inflation, and monetary-tightening shock of 2022, and the third with post-election policy and interest-rate repricing. These results suggest that SJM-Frequency responds more strongly to persistent macro-financial transitions than to isolated shocks.

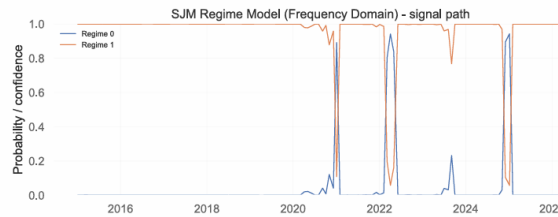


Figure 4: Out-of-sample SJM-Frequency regime probabilities. Compared with SJM-Time, the frequency-domain model identifies additional Regime 0 episodes after COVID, especially around macro-financial repricing periods.

The one-step-ahead forecast diagnostics further support this interpretation as shown in Table 5. SJM-Time achieves higher raw accuracy, 98.5%, because Regime 1 dominates

the test period. SJM-Frequency has slightly lower raw accuracy, 96.3%. Its balanced accuracy is relatively higher, 77.4% compared with 75.0% for SJM-Time. This suggests that the frequency-domain representation sacrifices a small amount of overall accuracy in exchange for better recognition of rare regime episodes.

Table 5: Out-of-Sample SJM Forecast Classification Results

Specification	Accuracy	Balanced Accuracy	M. Precision	M. Recall	M. F1
SJM-Time	98.5%	75.0%	99.3%	75.0%	83.0%
SJM-Frequency	96.3%	77.4%	90.1%	77.4%	82.4%

Overall, the two specifications capture different forms of unusual market conditions. SJM-Time identifies the acute COVID-19 stress episode, whereas SJM-Frequency identifies a broader set of persistent macro-financial repricing periods.

5.1.3 LPPLS Results

Unlike the HMM and the SJM, the LPPLS component does not partition the test period into persistent market regimes. It produces an episodic positive bubble-confidence indicator for each ETF, which is zero most of the time and spikes only when the recent price path is well described by faster-than-exponential growth with accelerating log-periodic oscillations. Because the indicator is estimated from daily log prices alone, the signal reported here is identical in the time-domain and frequency-domain experiments; the two variants differ only in the StaticMVO benchmark to which the overlay is applied.

Over the out-of-sample period from January 2015 to April 2026, the de-risking overlay was active at 64 of the 136 monthly rebalance dates in the time-domain implementation and 57 of 136 in the frequency-domain implementation, typically trimming one to three ETFs at a time. The cash sleeve created by de-risking averaged 1.3% of the portfolio in the time domain (1.1% in the frequency domain) and peaked at 21% in June 2025, so the strategy remained close to fully invested throughout the test period.

The flagged episodes line up with well-known market events, which we regard as a qualitative validation of the signal. Table 6 lists the main out-of-sample episodes in which an asset reached a positive bubble confidence of at least 0.5. The strongest cluster occurs between December 2017 and February 2018, when SPY, QQQ, XLK, and XLB were flagged simultaneously during the late-2017 melt-up, with risk multipliers reaching their floor values immediately before the February 2018 volatility spike. Other notable episodes include aggregate bonds (AGG) during the 2016 rate rally, investment-grade credit (LQD) in the late summer of 2019, gold (GLD) in August 2020 shortly after its all-time high and again in 2025 and early 2026, emerging markets (EEM) in early 2021 near the post-pandemic peak, health care (XLV) in the summer of 2021, and SPY in December 2024 ahead of the early-2025 correction. In the discovery-period history, the indicator also flags the 2006 to 2007 run-ups in Eurozone equities and emerging markets that preceded the Global Financial Crisis, consistent with the historical bubble cases documented in Sornette (2003) and Zhang et al. (2016).

Table 6: Main Out-of-Sample LPPLS Bubble Episodes (positive confidence ≥ 0.5)

Period	Assets Flagged	Market Context
Apr–Sep 2016	AGG, LQD	Bond rally toward historic low yields
Dec 2017–Feb 2018	SPY, QQQ, XLK, XLB	Late-2017 equity melt-up before the February 2018 volatility spike
Aug–Sep 2019	LQD, AGG	Rate rally and credit spread compression
Jul–Sep 2020	GLD, AGG	Gold all-time high after pandemic easing
Feb–Mar 2021	EEM	Post-pandemic emerging-market peak
Jul–Sep 2021	XLV	Health-care rally before late-2021 peak
Dec 2024–Jan 2025	SPY, GLD	Post-election rally before early-2025 correction
May–Jun 2025	GLD	Gold rally; largest overlay de-risking
Oct–Dec 2025	LQD, EEM	Credit and emerging-market run-ups
Jan–Mar 2026	GLD, EEM, EZU	Ongoing precious-metal and non-U.S. equity rallies

The behavior of the indicator is therefore complementary to the regime classifiers: it is silent during most of the sample, including sustained but orderly advances, and concentrates its signals in short windows of accelerating price growth. This selectivity is what allows the overlay to modify the benchmark allocation only when crash risk is elevated, rather than maintaining a persistent defensive tilt.

5.2 Portfolio Performance Results

5.2.1 Hidden Markov Model Portfolio Results

Table 7 and 8 compares each HMM-conditioned portfolio with its corresponding StaticMVO benchmark using the same asset universe, long-only constraints, maximum asset weight of 30%, and daily realized return convention. Within-domain comparisons are emphasized because the time- and frequency-domain benchmarks have slightly different feature availability and rebalance calendars.

Table 7: HMM Portfolio Performance

Specification	Ann. Return	Ann. Vol.	Sharpe	Sortino	Calmar
HMM-Time Regime-Aware	8.27%	14.67%	0.564	0.708	0.258
HMM-Time Static	13.98%	15.96%	0.876	1.115	0.538
HMM-Frequency Regime-Aware	12.03%	16.54%	0.727	0.914	0.416
HMM-Frequency Static	14.22%	15.98%	0.890	1.136	0.554

Table 8: HMM Downside Risk and Turnover

Specification	Max. Drawdown	VaR 95%	CVaR 95%	Avg. Turnover
HMM-Time Regime-Aware	-32.03%	-1.50%	-2.32%	21.09%
HMM-Time Static	-26.00%	-1.52%	-2.38%	4.26%
HMM-Frequency Regime-Aware	-28.92%	-1.51%	-2.42%	14.03%
HMM-Frequency Static	-25.66%	-1.52%	-2.38%	3.83%

Neither HMM-conditioned strategy outperformed its corresponding StaticMVO benchmark. HMM-Time reduced annualized volatility modestly, but experienced lower return, weaker Sharpe and Sortino ratios, a larger maximum drawdown, and substantially higher turnover. Its slightly less negative VaR and CVaR estimates were therefore insufficient to offset its weaker overall performance.

HMM-Frequency performed better than HMM-Time in annualized return, Sharpe ratio, Calmar ratio, and turnover. Nevertheless, it remained below its corresponding static benchmark and experienced a larger maximum drawdown. Frequency-domain decomposition, therefore, narrowed the performance gap but did not produce economic dominance over StaticMVO.

5.2.2 Statistical Jump Model Portfolio Results

As shown in Table 9 and 10, the SJM portfolio results compare RegimeAwareMVO against StaticMVO with same settings as noted in that of HMM, and similarly across domains.

For the time-domain specification, SJM-Time-RegimeAwareMVO does not improve overall return performance relative to StaticMVO. It achieves an annualized return of 13.23%, compared with 13.98%. Annualized volatility increases from 15.96% to 16.71%, while the Sharpe ratio declines from 0.876 to 0.791. However, the regime-aware strategy exhibits a smaller maximum drawdown of -23.97%, compared with -26.00% for StaticMVO, resulting in a slightly higher Calmar ratio of 0.552 versus 0.538. Average turnover also increases from 4.26% to 7.04%, indicating more frequent portfolio adjustments under the regime-aware allocation.

The frequency-domain specification presents a different performance profile. SJM-Frequency-RegimeAwareMVO demonstrates an annualized return increases from 14.22% to 14.94%. The higher return is accompanied by higher annualized volatility of 17.36%, compared with 15.98% for the static benchmark. Consequently, the Sharpe ratio decreases slightly from 0.890 to 0.860. Maximum drawdown also becomes larger, increasing from -25.66% to -27.83%, while average turnover rises from 3.83% to 8.14%.

Table 9: SJM Portfolio Performance

Specification	Ann. Return	Ann. Vol.	Sharpe	Sortino	Calmar
SJM-Time Regime-Aware	13.23%	16.71%	0.791	1.009	0.552
SJM-Time Static	13.98%	15.96%	0.876	1.115	0.538
SJM-Frequency Regime-Aware	14.94%	17.36%	0.860	1.084	0.537
SJM-Frequency Static	14.22%	15.98%	0.890	1.136	0.554

Table 10: SJM Downside Risk and Turnover

Specification	Max. Drawdown	VaR 95%	CVaR 95%	Avg. Turnover
SJM-Time Regime-Aware	-23.97%	-1.66%	-2.56%	7.04%
SJM-Time Static	-26.00%	-1.52%	-2.38%	4.26%
SJM-Frequency Regime-Aware	-27.83%	-1.61%	-2.58%	8.14%
SJM-Frequency Static	-25.66%	-1.52%	-2.38%	3.83%

Overall, the two SJM specifications exhibit different performance characteristics. The time-domain strategy provides modest improvement in downside protection through a

smaller maximum drawdown, but this is accompanied by lower returns and weaker risk-adjusted performance. In contrast, the frequency-domain strategy delivers higher cumulative and annualized returns than its corresponding static benchmark, although these gains are achieved alongside higher volatility, larger drawdowns, and more active portfolio rebalancing.

5.2.3 LPPLS Portfolio Results

As shown in Table 11 and 12, the LPPLS portfolio results compare RegimeAwareMVO against StaticMVO with same settings as noted in that of HMM and SJM. Because the overlay multiplies the StaticMVO weights directly, any performance difference is attributable solely to the bubble-signal de-risking.

In the time domain, the overlay improves every reported dimension simultaneously. It produces a cumulative return of 341.3%, compared with 335.1% for StaticMVO, and an annualized return of 14.13% versus 13.98%. Annualized volatility is lower, 15.75% versus 15.96%, so the Sharpe ratio rises from 0.876 to 0.897. The maximum drawdown improves slightly, from -26.00% to -25.89% , and the Calmar ratio rises from 0.538 to 0.546.

The frequency-domain comparison shows a larger improvement. The overlay achieves a cumulative return of 364.5%, compared with 345.2% for StaticMVO, and an annualized return of 14.65% versus 14.22%. Volatility falls from 15.98% to 15.46%, lifting the Sharpe ratio from 0.890 to 0.948. The maximum drawdown is essentially unchanged, -25.63% versus -25.66% , while the Calmar ratio improves from 0.554 to 0.572.

Table 11: LPPLS Overlay Portfolio Performance

Specification	Ann. Return	Ann. Vol.	Sharpe	Sortino	Calmar
LPPLS-Time Overlay	14.13%	15.75%	0.897	1.141	0.546
LPPLS-Time Static	13.98%	15.96%	0.876	1.115	0.538
LPPLS-Frequency Overlay	14.65%	15.46%	0.948	1.230	0.572
LPPLS-Frequency Static	14.22%	15.98%	0.890	1.136	0.554

Table 12: LPPLS Overlay Downside Risk and Turnover

Specification	Max. Drawdown	VaR 95%	CVaR 95%	Avg. Turnover
LPPLS-Time Overlay	-25.89%	-1.48%	-2.33%	6.10%
LPPLS-Time Static	-26.00%	-1.52%	-2.38%	4.26%
LPPLS-Frequency Overlay	-25.63%	-1.47%	-2.30%	5.38%
LPPLS-Frequency Static	-25.66%	-1.52%	-2.38%	3.83%

Overall, the LPPLS overlay outperforms StaticMVO across most reported metrics in both domains. The frequency-domain comparison shows the larger improvement, while average turnover increases from 4.26% to 6.10% in the time-domain implementation and from 3.83% to 5.38% in the frequency-domain implementation.

6 Analysis and Discussion

6.1 Cross-Model Interpretation

The empirical results show that regime identification, regime forecasting, and portfolio improvement are related but distinct objectives. Across the six model variants, stronger statistical regime signals do not automatically translate into stronger portfolio performance. This distinction is central to interpreting the framework. HMM and SJM attempt to infer broad market states and then translate those states into conditional return and covariance inputs for RegimeAwareMVO. LPPLS, by contrast, does not classify the market into persistent regimes; it applies a sparse asset-level de-risking overlay to the StaticMVO benchmark when bubble confidence is elevated.

The results suggest three different roles for the models. HMM is useful as a probabilistic market-state monitoring tool, but its portfolio results are weaker than StaticMVO in both domains. SJM provides a more selective regime mechanism: the time-domain version improves maximum drawdown, while the frequency-domain version improves annualized return, but neither specification dominates StaticMVO across all risk-adjusted metrics. LPPLS provides the clearest portfolio benefit in the current experiment because it remains close to the benchmark and intervenes only during short windows of elevated bubble confidence.

6.2 HMM: Stronger State Forecasting Does Not Guarantee Portfolio Dominance

The HMM findings highlight the difference between forecasting a latent state and generating economically valuable portfolio inputs. The frequency-domain HMM produced more persistent discovery-period regimes than the time-domain HMM, with only 10 transitions and an average duration of 34.8 weeks, compared with 34 transitions and an average duration of 13.0 weeks for HMM-Time. The frequency-domain specification also achieved stronger out-of-sample forecast diagnostics in the reported out-of-sample results, with accuracy of 98.53%, balanced accuracy of 96.43%, and macro F1 of 97.69%.

However, this statistical improvement did not translate into portfolio dominance. HMM-Time generated an annualized return of 8.27%, Sharpe ratio of 0.564, Calmar ratio of 0.258, and maximum drawdown of -32.03%, all weaker than its StaticMVO benchmark, which achieved 13.98% annualized return, Sharpe ratio of 0.876, Calmar ratio of 0.538, and maximum drawdown of -26.00%. HMM-Frequency narrowed the gap, with annualized return of 12.03% and Sharpe ratio of 0.727, but still remained below its corresponding StaticMVO benchmark, which achieved 14.22% annualized return and Sharpe ratio of 0.890. These results demonstrate that accurate prediction of a latent market state does not necessarily imply accurate estimation of the conditional expected returns and covariance matrices required for mean–variance portfolio optimization.

Several mechanisms may explain this result. First, the HMM regimes are latent statistical constructs rather than directly observed economic states. High forecasting accuracy therefore measures consistency with the model’s own state structure instead of correctness against an externally observable market-regime label. Moreover, the less frequently observed regime, particularly in the frequency-domain specification, is estimated from relatively few observations, increasing uncertainty in the corresponding conditional return and covariance estimates. Second, regime-aware mean–variance optimization is

highly sensitive to expected-return estimation error. Even small changes in probability-weighted conditional means can produce disproportionately large changes in optimized portfolio weights. Third, the HMM-conditioned strategies generated substantially higher turnover than StaticMVO. HMM-Time had average turnover of 21.09%, compared with 4.26% for StaticMVO, while HMM-Frequency had average turnover of 14.03%, compared with 3.83% for its benchmark. Because transaction costs are not deducted from the reported returns, the results may overstate the net performance of the regime-aware portfolios. Incorporating even modest trading costs would likely reduce their relative economic attractiveness further.

6.3 SJM: Selective Regime Detection with Mixed Portfolio Benefits

The SJM results provide a useful contrast to HMM. Both SJM specifications selected a two-regime structure with jump penalty $\lambda = 10$, but they identified different types of unusual market conditions. SJM-Time classified only two of the 136 monthly out-of-sample rebalance dates as Regime 0, both during the immediate COVID-19 stress period. SJM-Frequency classified six rebalance dates as Regime 0, including January 2021, March–May 2022, and December 2024–January 2025. This suggests that the frequency-domain representation makes SJM more responsive to persistent macro-financial repricing episodes, whereas the time-domain model reacts mainly to acute market stress.

The portfolio results are mixed rather than uniformly positive. SJM-Time did not improve annualized return or Sharpe ratio relative to StaticMVO. Its annualized return was 13.23%, compared with 13.98% for StaticMVO, and its Sharpe ratio declined from 0.876 to 0.791. However, it reduced maximum drawdown from -26.00% to -23.97% and slightly improved the Calmar ratio from 0.538 to 0.552. This indicates that SJM-Time provided some downside protection, although at the cost of lower return, higher volatility, weaker Sortino ratio, and higher turnover.

SJM-Frequency showed a different trade-off. It increased annualized return from 14.22% to 14.94%, but annualized volatility rose from 15.98% to 17.36%. Its Sharpe ratio declined from 0.890 to 0.860, its maximum drawdown worsened from -25.66% to -27.83%, and average turnover increased from 3.83% to 8.14%. Thus, the frequency-domain SJM improved raw return but did not improve broad risk-adjusted performance.

Overall, SJM appears more economically useful than HMM in specific dimensions, but not as a uniformly superior allocation engine. If the objective is drawdown mitigation, SJM-Time provides some evidence of benefit. If the objective is higher annualized return, SJM-Frequency performs better. If the objective is broad dominance across return, volatility, drawdown, tail risk, and turnover, neither SJM specification clearly dominates StaticMVO.

6.4 LPPLS: Sparse Overlay with the Clearest Portfolio Benefit

The LPPLS overlay contributes to the framework in a fundamentally different way from HMM and SJM. Rather than classifying the market into persistent regimes and conditioning portfolio optimization on estimated regime probabilities, LPPLS operates as a sparse asset-level risk overlay. It directly scales the StaticMVO portfolio weights according to bubble confidence and intervenes only when crash risk is estimated to be elevated. Consequently, the portfolio remains close to the benchmark during normal market conditions,

limiting both its potential outperformance and its potential underperformance relative to StaticMVO. Its value therefore lies in delivering consistent incremental improvement rather than making large tactical allocation shifts.

In the current results, LPPLS provides the strongest portfolio evidence among the three model families. In the time-domain comparison, LPPLSOverlayMVO increased annualized return from 13.98% to 14.13%, reduced annualized volatility from 15.96% to 15.75%, improved the Sharpe ratio from 0.876 to 0.897, improved the Sortino ratio from 1.115 to 1.141, and slightly reduced maximum drawdown from -26.00% to -25.89%. In the frequency-domain comparison, the improvement was larger: annualized return increased from 14.22% to 14.65%, annualized volatility declined from 15.98% to 15.46%, the Sharpe ratio increased from 0.890 to 0.948, and the Sortino ratio increased from 1.136 to 1.230.

The downside-risk metrics further reinforce this conclusion. LPPLS-Time improved VaR_{95} from -1.52% to -1.48% and CVaR_{95} from -2.38% to -2.33%, while LPPLS-Frequency improved VaR_{95} from -1.52% to -1.47% and CVaR_{95} from -2.38% to -2.30%. Unlike the regime-classification approaches, LPPLS improves multiple portfolio-performance dimensions simultaneously. Across both domains, annualized return, volatility, Sharpe ratio, Sortino ratio, VaR, and CVaR all improve, while the increase in average turnover remains modest, rising from 4.26% to 6.10% in the time-domain implementation and from 3.83% to 5.38% in the frequency-domain implementation. This pattern is consistent with the objective of providing incremental downside-risk reduction without substantially altering the underlying strategic allocation.

The source of these improvements is consistent with the intended role of LPPLS as a crash-risk overlay. By making small, targeted reductions in exposure only when bubble confidence is elevated, the strategy sacrifices little upside participation during normal markets while reducing exposure during periods of elevated crash risk. The flagged episodes, including the late-2017 equity melt-up before the February 2018 volatility spike, provide qualitative support for interpreting the LPPLS signal as an episodic early-warning indicator rather than a persistent market-regime classifier. For a wealth-management setting, this overlay is also easier to communicate and implement than a fully regime-conditioned allocation because it addresses a narrower and more intuitive decision: whether temporarily to reduce exposure to assets exhibiting unstable price dynamics instead of continuously reallocating the entire portfolio according to inferred market regimes.

6.5 Practical Implications, Limitations, and Future Work

The empirical results suggest that regime awareness should be viewed as a complement to, rather than a replacement for, conventional portfolio optimization. The three model families contribute different types of information: HMM provides probabilistic market-state monitoring, SJM identifies selective stress and macro-financial repricing episodes, and LPPLS acts as a sparse asset-level crash-risk overlay. Accordingly, their value should be assessed according to the investment objective they support rather than whether every model consistently outperforms the StaticMVO benchmark. Evaluating all approaches under a common investment universe, optimization framework, and benchmark makes it possible to isolate the incremental contribution of each methodology and provides a clearer interpretation of their respective economic roles.

Several limitations should be acknowledged. First, although the backtest is implemented in a walk-forward manner, the evaluation period was unavoidably repeatedly

reviewed during model development. The results should therefore be interpreted as pseudo out-of-sample evidence rather than as a fully blind holdout test. Second, transaction costs are not deducted from the reported returns. Although the assets are liquid ETFs, even modest trading costs may materially affect the higher-turnover HMM and SJM strategies. Net-of-cost performance should therefore be evaluated as a robustness check. Third, regime-aware mean–variance optimization remains sensitive to estimation error in regime-specific expected returns and covariance matrices, especially when rare regimes contain relatively few observations. For LPPLS, the overlay cannot protect against crashes that are not preceded by super-exponential price growth, such as unexpected exogenous shocks.

Several directions remain for future research. A systematic sensitivity analysis of the LPPLS overlay parameters, including the de-risking threshold, sigmoid steepness, and minimum allocation multiplier, would provide a more comprehensive understanding of the overlay’s robustness. In the current dashboard implementation, investor risk preference is controlled through the shared mean–variance risk-aversion parameter (γ), while the LPPLS overlay parameters remain fixed. Finally, an LLM-based narrative verification layer could be incorporated to retrieve and summarize contemporaneous news, central-bank communications, earnings reports, and geopolitical events around detected regimes or LPPLS warning signals. Such qualitative information would complement the quantitative models by improving the interpretability of detected market events and helping distinguish economically meaningful warnings from false positives.

7 Conclusion

This project developed and evaluated a unified regime-aware portfolio management framework integrating feature preprocessing, market-state identification, regime forecasting, crash-risk detection, and portfolio construction within a single investment pipeline. Hidden Markov Models (HMM), Statistical Jump Models (SJM), and the Log-Periodic Power Law Singularity (LPPLS) model were implemented under a common experimental setting using the same investment universe, portfolio optimization framework, benchmark strategy, and walk-forward evaluation procedure. Both time-domain and frequency-domain feature representations were considered in order to examine whether multi-scale information could improve regime identification and ultimately lead to more informed portfolio decisions.

The empirical results demonstrate that statistical regime identification and economic portfolio performance are related but fundamentally different objectives. Frequency-domain feature representations generally produced more persistent regimes and stronger forecasting performance for both HMM and SJM, indicating that decomposing market information across different time horizons can improve the stability of regime identification. However, stronger statistical performance did not consistently translate into superior portfolio performance. In particular, the HMM-based strategies illustrate that accurate forecasting of latent market states does not necessarily imply accurate estimation of the conditional expected returns and covariance matrices required for mean–variance optimization. Although SJM exhibited greater economic value in certain dimensions, such as drawdown reduction or return enhancement depending on the specification, neither regime-classification approach consistently outperformed the StaticMVO benchmark across all portfolio metrics.

Among the evaluated methodologies, the LPPLS overlay provided the clearest portfolio improvement. By acting as a sparse asset-level crash-risk overlay rather than a full regime-classification model, LPPLS remained close to the benchmark allocation while delivering consistent improvements across return, volatility, downside-risk, and risk-adjusted performance measures with only a modest increase in portfolio turnover. These findings suggest that crash-risk overlays and regime-classification models play complementary rather than competing roles within a regime-aware investment framework. While HMM and SJM contribute information about evolving market conditions, LPPLS contributes targeted downside-risk management that enhances an existing strategic allocation.

Beyond the individual model comparisons, the primary contribution of this project lies in the construction of a unified evaluation framework for regime-aware investing. By comparing multiple methodologies under identical investment assumptions, optimization constraints, and evaluation procedures, the framework isolates the incremental contribution of different sources of market information and provides a transparent basis for assessing their economic value. This design also establishes a flexible foundation upon which additional regime-detection models, forecasting techniques, or portfolio-construction methods can be incorporated and evaluated consistently.

Overall, the results support the view that regime awareness should be regarded as an enhancement to conventional portfolio optimization rather than a replacement for it. Different quantitative models capture different aspects of market dynamics, and no single methodology is sufficient under all market conditions. A practical regime-aware investment system is therefore likely to benefit from combining complementary sources of information instead of relying on a single regime-identification approach. Future developments, including adaptive overlay parameter selection, richer frequency-domain representations, alternative portfolio optimization techniques, and LLM-assisted narrative verification of detected market events, may further improve the robustness, interpretability, and practical applicability of regime-aware portfolio management systems.

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